Music Minimal Skip Optimizer

STAT 4830 Slides Week 9

Problem Statement

Skipping Songs

Skip behavior is core engagement sign for music platforms.

Skips can indicate:

- Shorter listening sessions
- Lower satisfaction
- Reduced retention

Objective

Given a fixed set of candidate tracks, find the best permutation to minimize cumulative predicted skip probability

Week 6 focus: optimize sequence quality in later session positions using a trained skip model

Impact

Ordering is a real product decision (Daily Mix, Radio, Discover Weekly)

Better ordering improves listening flow without changing the underlying songs.

Constraints

Constraints (Week 6)

- Model depends on richer features; preprocessing is heavier
- Tracks mostly independently treated when scoring, so transition quality not fully optimized
- Search still greedy, so can’t guarantee globally optimal sequence order
- Validation remains offline (session-based split), so no direct causal proof from user behavior.

Data Updates

Spotify Sequential Skip Prediction (WSDM Cup 2018)

Data Updates

- Merged Spotify session logs + track metadata/audio features
- 167,880 merged rows
- 8,000 train / 2,000 test sessions

Feature Engineering

- Where the user is in the session and time context (*session_progress*, day/time/weekend)
- How smooth the transition is (*delta_**, acoustic similarity)
- Cumulative user behavior so far (*cum_skips*, *skip_rate_so_far*, pauses, tempo deviation)

Why it's suitable

- Moved from mostly heuristic signals to a context-aware feature space

Transformer Model Methodology

Goal: Build Transformer model with input of compound tokens which consist of multiple features, and output of two heads of SKIP or SONG (no skip).

- For song token: `E_song(title) + E_genre(genre) + E_pos`
- For skip token: `E_skip + E_pos`

Output heads:

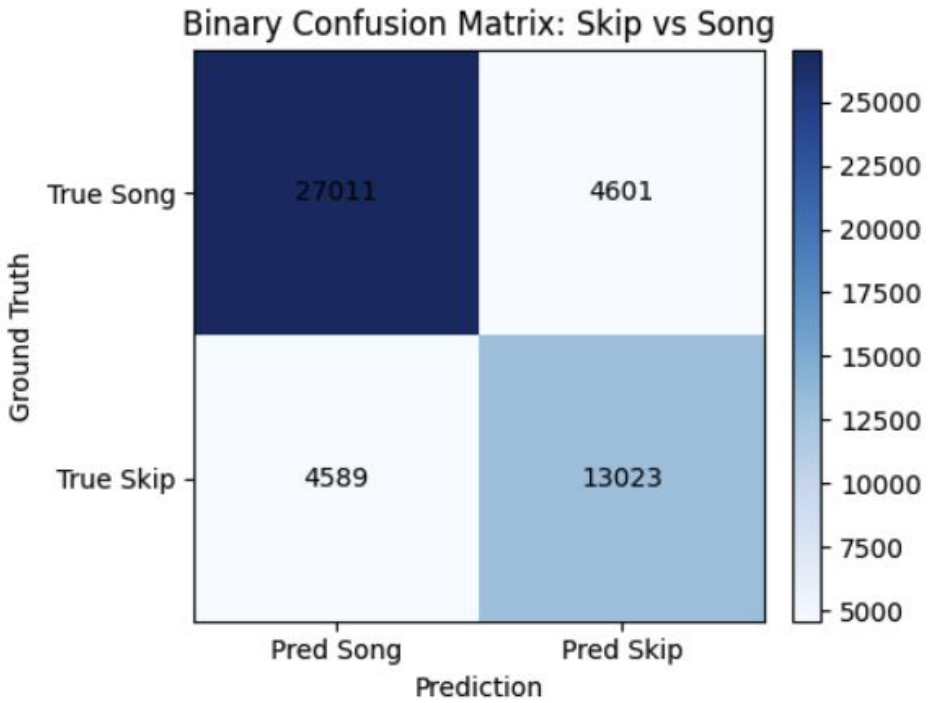
- `song_head` : next-song token distribution (over song tokens only)
- `skip_head` : probability next token is `SKIP`

Final decoded next token:

- if `p(skip) >= threshold` -> `SKIP`
- else -> argmax song token

Transformer Results

- The skip precision is 74%, and the recall is 74% (vs. 73% and 63% for the baseline probability model).
- Optimized threshold for skipping is 0.6879.
- Performance seems reasonably better than baseline model, however more features/complexity can be beneficial.



```
CompoundTokenTransformer(
  (song_emb): Embedding(14046, 128)
  (genre_emb): Embedding(2, 128)
  (pos_emb): Embedding(40, 128)
  (dropout): Dropout(p=0.1, inplace=False)
  (encoder): TransformerEncoder(
    (layers): ModuleList(
      (0-1): 2 x TransformerEncoderLayer(
        (self_attn): MultiheadAttention(
          (out_proj): NonDynamicallyQuantizableLinear(in_features=128, out_features=128, bias=Tru
e)
        )
        (linear1): Linear(in_features=128, out_features=256, bias=True)
        (dropout): Dropout(p=0.1, inplace=False)
        (linear2): Linear(in_features=256, out_features=128, bias=True)
        (norm1): LayerNorm((128,), eps=1e-05, elementwise_affine=True)
        (norm2): LayerNorm((128,), eps=1e-05, elementwise_affine=True)
        (dropout1): Dropout(p=0.1, inplace=False)
        (dropout2): Dropout(p=0.1, inplace=False)
      )
    )
  )
  (song_head): Linear(in_features=128, out_features=14045, bias=True)
  (skip_head): Linear(in_features=128, out_features=1, bias=True)
)
```

Next Steps

- Add in the other included features such as danceability, energy, flatness, tempo etc. into the embedding
- Develop optimal sequence algorithm -> use to generate playlists
- Handle long songs explicitly
    - Frequency-aware training, class-balanced sampling, or two-tower item embeddings.
    - Most performance loss is typically on rare items.

Thank You!